

NORTHERN UNIVERSITY BANGLADESH



E-Learning Management System

This project report has been submitted to the Department of Computer Science and Engineering, Northern University Bangladesh, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

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DECLARATION

We do hereby solemnly declare that the project report entitled, “**E-Learning Management System**” submitted to Northern University Bangladesh is a record of an original work done by us under the guidance of our respected supervisor **Avishek Das**, Lecturer, Department of Computer Science and Engineering (CSE), Faculty of Engineering, Northern University Bangladesh.

We, hereby guarantee that the work that have been presented here do not breach any existing copyright. We further undertake to indemnify the University against any loss or damage arising from breach of the foregoing obligations.

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CERTIFICATE

This is to certify that this Project Report titled, “**E-Learning Management System**” is the confide record of the project work done by Amirul Islam, Md Saroar Seheen and Sheikh Shadikur Rahman for partial fulfillment of requirements of the degree of BSc in Computer Science and Engineering from Northern University Bangladesh.

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ABSTRACT

This project presents the development and implementation of E-Learning Management System, a comprehensive web-based platform designed to modernize university education through digital innovation. The system addresses critical gaps in traditional university communication and learning methods by providing a centralized, interactive platform that integrates modern web technologies with educational best practices. The E-Learning Management System platform is built using a modern full-stack architecture comprising React front-end, Node.js back-end, and MySQL database, ensuring scalability, security, and user experience excellence. The system successfully implements core features including user authentication and role-based access control, comprehensive course management, video streaming capabilities supporting files up to 500MB, interactive quiz and assessment systems, and real-time notification services. Additionally, the platform incorporates advanced features such as contact form management with email notifications and administrative dashboard functionality for comprehensive system oversight. The development methodology follows an iterative, agile approach that emphasizes continuous improvement and stakeholder collaboration. Performance testing demonstrates that the platform maintains responsive user experience under moderate load conditions, with page load times consistently under three seconds and efficient handling of concurrent user sessions. User interface testing reveals high usability scores across different user roles, validating the effectiveness of our user-centered design approach.

Glossary

API – Application Programming Interface for building and integrating software

Authentication – Verifying user identity

Authorization – Determining user permissions

Back-end – Server-side portion of web applications

CRUD – Create, Read, Update, Delete database operations

CSS – Cascading Style Sheets for web page styling

Database – Organized collection of structured data

Frontend – Client-side portion users interact with

Full-Stack – Development covering both front-end and back-end

HTML – Hypertext Markup Language for web documents

HTTP/HTTPS – Protocols for web data transmission (secure)

JWT – JSON Web Token for secure authentication

LMS – Learning Management System for educational content

Middleware – Software bridge between applications and systems

MySQL – Relational database management system

Node.js – JavaScript runtime environment for server-side development

ORM – Object-Relational Mapping for database operations

Prisma – Modern database toolkit for Node.js

React – JavaScript library for building user interfaces

REST – Representational State Transfer architecture style

SQL – Structured Query Language for database management

Type-script – JavaScript with static typing

UI/UX – User Interface and User Experience design

XSS/CSRF – Security vulnerabilities (Cross-Site Scripting/Request Forgery)

JWT Token – Secure authentication token

Nodemailer – Email sending module for Node.js

Multer – File upload middleware

SweetAlert2 – Beautiful JavaScript popup replacement

Tailwind CSS – Utility-first CSS framework

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CHAPTER 1

Introduction

1.1 Objectives

The E-Learning Management System project represents a sophisticated, full-stack digital learning management system designed specifically for university education. The system serves as a centralized hub that integrates multiple educational components:

Core Educational Features:

- Course Management System: Comprehensive course creation, management, and delivery
- Multi-Media Content Delivery: Video streaming, document sharing, and interactive content
- Assessment & Evaluation: Quiz creation, automated grading, and performance tracking
- Student Progress Monitoring: Detailed analytics and progress tracking
- Payment Integration: Secure payment processing with multiple gateway support

User Role Management:

- Students: Access to courses, quizzes, evaluations, and learning materials
- Teachers: Course creation, content management, and student assessment tools
- Administrators: System-wide management, user approvals, and analytics

1.2 Motivations

Our motivation stems from recognizing the significant gaps in current university communication and learning methods. Traditional approaches often lack the flexibility and engagement that modern students expect, while existing digital solutions may not fully address the unique needs of university ecosystems. We are driven by the desire to create an innovative platform that fosters community engagement, enables personalized learning experiences, and leverages cutting-edge technology to enhance educational outcomes.

Key motivations:

- **Bridging Gaps in Communication:** Current communication between students, teachers, and administrators is fragmented, often spread across emails, social media groups, and physical notice boards. A centralized E-Learning Management System system eliminates these inefficiencies by offering a unified communication hub.
- **Enhancing Engagement and Retention:** Traditional lectures may not always capture student attention. Digital features like interactive quizzes, gamified modules, and real-time feedback significantly boost engagement and knowledge retention.
- **Addressing Limitations of Existing Platforms:** While platforms such as Moodle, Blackboard, and Canvas exist, they often have limitations in terms of user experience, customization for local university needs, and integration with institutional systems.
- **Adapting to Technological Shifts:** The rise of mobile learning, cloud computing, artificial intelligence, and data analytics creates opportunities for smarter, more effective education systems. Our project leverages these technologies to create a future-proof platform.
- **Promoting Inclusivity and Accessibility:** Education must be available to all learners regardless of location, financial constraints, or physical limitations. An E-Learning Management System can provide remote access, assistive technologies, and multiple learning formats to promote inclusivity.

In short, our motivation is driven by the desire to modernize higher education and build a system that aligns with the expectations of today's students, faculty, and administrators.

1.3 Background Studies

Extensive Frontend Technology Stack:

- **React 19.1.0:** Modern React with latest features and performance optimizations
- **TypeScript 4.9.5:** Type-safe development for better code quality
- **Tailwind CSS 3.4.17:** Utility-first CSS framework for responsive design
- **React Router 7.6.3:** Advanced routing with protected routes and role-based access
- **Lucide React:** Modern icon library for consistent UI elements

Backend Technology Stack:

- Node.js with Express: Robust server-side framework
- TypeScript 5.8.3: Backend type safety and modern JavaScript features
- Prisma ORM: Modern database toolkit with MySQL support
- JWT Authentication: Secure token-based authentication system
- Multer: File upload handling for multimedia content

Database Architecture:

- MySQL Database: Reliable relational database system
- Prisma Migrations: Version-controlled database schema management
- Comprehensive Data Models: 15+ interconnected models for complete system functionality

1.4 Problem Statement

The current university education system faces several critical challenges that our E-Learning Management System platform aims to solve. There is a lack of centralized access to educational resources, limited personalization in learning experiences, and insufficient real-time communication between students, teachers, and administrators. Traditional methods often result in information silos, reduced engagement, and missed opportunities for collaborative learning and community building.

Centralized Resource Access:

- Fragmented Learning Materials: Current systems scatter content across multiple platforms
- Inconsistent User Experience: Different interfaces for different educational components
- Limited Integration: Poor connectivity between various educational tools and systems

Personalization & Engagement:

- Static Learning Paths: Lack of adaptive learning based on student performance
- Limited Interactive Elements: Minimal engagement through traditional methods
- Poor Progress Tracking: Inadequate monitoring of student advancement

Communication & Collaboration:

- Isolated Learning: Students often work in isolation without community support
- Delayed Feedback: Slow response times in traditional assessment systems
- Limited Teacher-Student Interaction: Restricted communication channels

1.5 Challenges

We Developing a comprehensive E-Learning Management System platform presents several significant challenges that we must address throughout the project lifecycle. Technical challenges include ensuring scalability for large user bases, implementing secure authentication and authorization systems, and creating an intuitive user interface that accommodates diverse user needs. Additionally, we face challenges in content management, data integration, and ensuring the platform remains accessible across various devices and network conditions.

Technical Challenges:

- Scalability Requirements: System must handle large numbers of concurrent users
- Security Implementation: Robust authentication, authorization, and data protection
- Performance Optimization: Fast response times for multimedia content delivery
- Cross-Platform Compatibility: Responsive design across various devices and browsers

Content Management Challenges:

- Multimedia Handling: Efficient storage and delivery of video, audio, and document content
- File Upload Management: Secure handling of various file types and sizes
- Content Organization: Logical structuring of educational materials

Integration Challenges:

- Payment Gateway Integration: Secure integration with multiple payment providers
- Third-party Service Integration: Seamless connection with external educational tools

- API Design: Comprehensive RESTful API architecture for frontend-backend communication

1.6 Project Organization

The E-Learning Management System project is organized into distinct phases that guide our development process from conception to deployment. We begin with comprehensive planning and requirements analysis, followed by system design and prototyping phases. The development phase involves iterative building and testing of core features, while the final phases focus on comprehensive testing, user training, and system deployment. Throughout this process, we maintain close collaboration with stakeholders and conduct regular reviews to ensure alignment with project goals.

Development Architecture:

- Modular Component Design: Reusable React components for consistent UI/UX
- Role-based Access Control: Secure routing and component access based on user roles
- Context-based State Management: React Context API for global state management
- Service Layer Architecture: Clean separation between UI and business logic

Database Schema Design:

- User Management System: Comprehensive user profiles with approval workflows
- Course & Content Management: Structured organization of educational materials
- Assessment & Evaluation: Complete quiz and evaluation tracking systems
- Payment & Enrollment: Secure financial transaction handling
- Communication Systems: Contact messaging and notification capabilities

Security Implementation:

- JWT Token Authentication: Secure session management
- Role-based Authorization: Granular access control based on user permissions
- Payment Verification: Secure payment processing and verification

- Data Validation: Input stigmatization and validation at multiple levels

1.6 Conclusion

The E-Learning Management System project represents a comprehensive solution that addresses the fundamental limitations of current university education systems. By implementing modern web technologies, robust security measures, and intuitive user interfaces, the platform creates an environment where:

- Students can access personalized learning experiences with real-time feedback
- Teachers can efficiently manage courses and assess student progress
- Administrators can monitor system performance and user engagement
- Institutions can provide modern, accessible, and engaging educational experiences

The system's modular architecture ensures scalability and maintainability, while its comprehensive feature set addresses the diverse needs of modern university education. Through this platform, universities can bridge the gap between traditional classroom learning and digital innovation, creating a more engaging, accessible, and effective educational environment.

The project demonstrates advanced software engineering practices, including:

- Type Safety: Full TypeScript implementation for both frontend and backend
- Modern React Patterns: Latest React features and best practices
- Database Design: Well-structured relational database with proper relationships
- API Architecture: RESTful API design with proper error handling
- Security: Comprehensive authentication and authorization systems

CHAPTER 2

Methodology

2.1 Previous Work

Our research began with an extensive and thorough review of existing learning management systems and educational technology platforms that have shaped the current landscape of digital education. We conducted a comprehensive analysis of prominent systems such as Canvas, Moodle, and Blackboard to understand their architectural patterns, feature sets, and implementation approaches. This analysis involved deep-diving into their source code repositories, documentation, and user experience patterns to identify both their strengths and limitations.

Canvas Learning Management System Analysis:

Canvas, developed by Instructure, represents one of the most modern and widely adopted LMS platforms in higher education. Our research focused on understanding its RESTful API architecture, which provides seamless integration capabilities with third-party tools and services. We analyzed how Canvas handles real-time notifications, mobile responsiveness, and its approach to content delivery. The platform's use of modern web technologies such as React and its microservices architecture provided valuable insights into building scalable educational applications.

Moodle Open-Source Platform Examination:

Moodle, being an open-source solution, offered us unique insights into community-driven development practices and plugin architecture. We extensively studied its modular design approach, which allows for extensive customization through plugins and themes. Our research included analyzing how Moodle handles different types of assessments, its database schema design for educational content, and its approach to user role management and permissions. The platform's extensive plugin ecosystem demonstrated the importance of extensibility in educational software.

Blackboard Learn Platform Investigation:

Blackboard Learn, one of the longest-standing LMS platforms, provided us with valuable insights into enterprise-level educational software architecture. We examined its approach to handling large-scale

deployments, its integration with student information systems, and its sophisticated assessment and grading workflows. The platform's focus on accessibility and compliance with educational standards informed our own development approach.

Academic Literature and Research Integration:

Beyond analyzing existing platforms, we conducted extensive research into academic literature on modern web application development, database design for educational systems, and user experience design principles. This research encompassed studies on cognitive load theory in digital learning environments, principles of effective educational interface design, and best practices for building responsive web applications that work across various devices and screen sizes.

Technical Research and Technology Selection:

Our research extended into the technical realm, where we analyzed various technology stacks and their suitability for educational applications. We examined the performance characteristics of different frontend frameworks, the scalability of various backend architectures, and the security implications of different authentication and authorization approaches. This research informed our decision to adopt React for the frontend, Node.js with Express for the backend, and MySQL with Prisma ORM for the database layer.

User Experience Research and Design Principles:

We conducted extensive research into user experience design principles specifically tailored for educational applications. This included studying how students interact with digital learning materials, how teachers manage course content, and how administrators oversee system operations. Our research revealed the importance of intuitive navigation, clear information hierarchy, and responsive design in maintaining user engagement and reducing cognitive load.

Security and Privacy Research:

Given the sensitive nature of educational data, we conducted thorough research into security best practices for educational software. This included studying data protection regulations, secure authentication mechanisms, and privacy-preserving approaches to user data management. Our research informed the implementation of JWT-based authentication, role-based access

2.2 Project Architecture

The E-Learning Management System platform follows a modern, scalable, and maintainable architecture that separates concerns and enables efficient development and maintenance. Our system employs a sophisticated three-tier architecture consisting of a React-based frontend for user interactions, a Node.js backend API for business logic and data processing, and a MySQL database for persistent data storage. This architecture supports real-time updates, responsive design, and seamless integration between different system components while maintaining security and performance.

Frontend Architecture - React-Based User Interface Layer:

Component-Based Architecture:

The frontend is built using React 19.1.0, leveraging the latest features and performance optimizations available in modern React. Our component architecture follows a hierarchical structure that promotes reusability and maintainability. Each component is designed to handle specific functionality while maintaining clear separation of concerns.

State Management and Context API:

We implement React Context API for global state management, creating a centralized approach to managing application state. The AuthContext manages user authentication state, user profile information, and authentication-related operations. The CartContext handles shopping cart functionality, course selection, and payment verification workflows. This approach ensures that state is consistently managed across the application while maintaining performance and avoiding unnecessary re-renders.

Routing and Navigation System:

The application implements a sophisticated routing system using React Router 7.6.3, which provides advanced routing capabilities including protected routes, role-based access control, and dynamic navigation. Our routing system automatically redirects users to appropriate dashboards based on their roles (Student, Teacher, or Administrator) and implements security measures to prevent unauthorized access to restricted areas.

Responsive Design and UI Framework:

We utilize Tailwind CSS 3.4.17 as our primary styling framework, implementing a utility-first approach that ensures consistent design patterns across the application. The responsive design system automatically adapts to various screen sizes and devices, providing an optimal user experience on desktop computers, tablets, and mobile devices.

Type Safety and Development Experience:

The entire frontend is built with TypeScript 4.9.5, providing comprehensive type safety and improving the development experience. This includes strict typing for API responses, component props, and application state, reducing runtime errors and improving code maintainability.

Backend Architecture - Node.js API Layer:

Express.js Framework and Middleware:

The backend is built using Express.js, a fast and minimalist web framework for Node.js. We implement a comprehensive middleware stack that includes security measures, request logging, error handling, and CORS configuration. The middleware architecture ensures that all requests are properly validated, authenticated, and processed according to security best practices.

API Design and RESTful Architecture:

Our API follows RESTful design principles, providing clear and consistent endpoints for all system operations. The API is organized into logical modules including authentication, user management, course management, content delivery, assessment systems, and payment processing. Each module implements proper error handling, input validation, and response formatting.

Security Implementation:

The backend implements multiple layers of security including JWT-based authentication, role-based access control, input sanitization, and secure headers using Helmet.js. We implement comprehensive CORS policies that allow secure communication between the frontend and backend while preventing unauthorized access from malicious origins.

File Upload and Media Handling:

The system includes sophisticated file upload capabilities using Multer middleware, supporting various file types including videos, documents, images, and other educational materials. We implement secure file storage with proper access controls and implement streaming capabilities for video content to ensure optimal performance.

Error Handling and Logging:

We implement a comprehensive error handling system that captures and logs all system errors, provides meaningful error messages to users, and maintains system stability. The logging system uses Morgan for HTTP request logging and custom error handlers for application-specific error scenarios.

Database Architecture - MySQL with Prisma ORM:

Relational Database Design:

The database is built using MySQL, providing a robust and reliable foundation for data storage. Our database schema is carefully designed to support complex educational workflows while maintaining data integrity and performance. The schema includes comprehensive models for users, courses, content, assessments, and financial transactions.

Prisma ORM Integration:

We utilize Prisma as our Object-Relational Mapping (ORM) tool, which provides type-safe database operations, automatic migrations, and efficient query building. Prisma's schema-first approach allows us to define our data models in a declarative way, automatically generating the necessary database tables and relationships.

Data Model Relationships:

Our database schema implements complex relationships between various entities including user enrollments, course content, assessment attempts, and payment transactions. These relationships are carefully designed to maintain referential integrity while supporting efficient querying and data retrieval operations.

Payment Gateway Integration:

We implement secure payment processing through multiple payment gateways, with primary support for SSLCommerz and custom payment methods. The payment system includes comprehensive transaction tracking, payment verification, and secure handling of financial data.

External Service Integration:

The system is designed to integrate seamlessly with external educational tools and services. This includes support for various content formats, integration with external authentication systems, and compatibility with standard educational data formats

Performance and Scalability Considerations:**Caching and Optimization:**

We implement various caching strategies including browser-level caching for static assets, API response caching, and efficient database query optimization. The system is designed to handle large numbers of concurrent users while maintaining optimal performance.

Load Balancing and Distribution:

The architecture supports horizontal scaling through load balancing and distributed deployment. The stateless nature of our API design allows for easy deployment across multiple server instances while maintaining consistent user experience.

Monitoring and Analytics:

We implement comprehensive monitoring and analytics capabilities that track system performance, user engagement, and system health. This includes performance metrics, error tracking, and user behavior analytics that inform system optimization and feature development.

This comprehensive architecture ensures that the E-Learning Management System platform is not only functional and secure but also scalable, maintainable, and capable of supporting the complex requirements of modern university education systems. The separation of concerns, modern technology stack, and comprehensive security measures create a robust foundation for delivering high-quality educational experiences to students, teachers, and administrators.

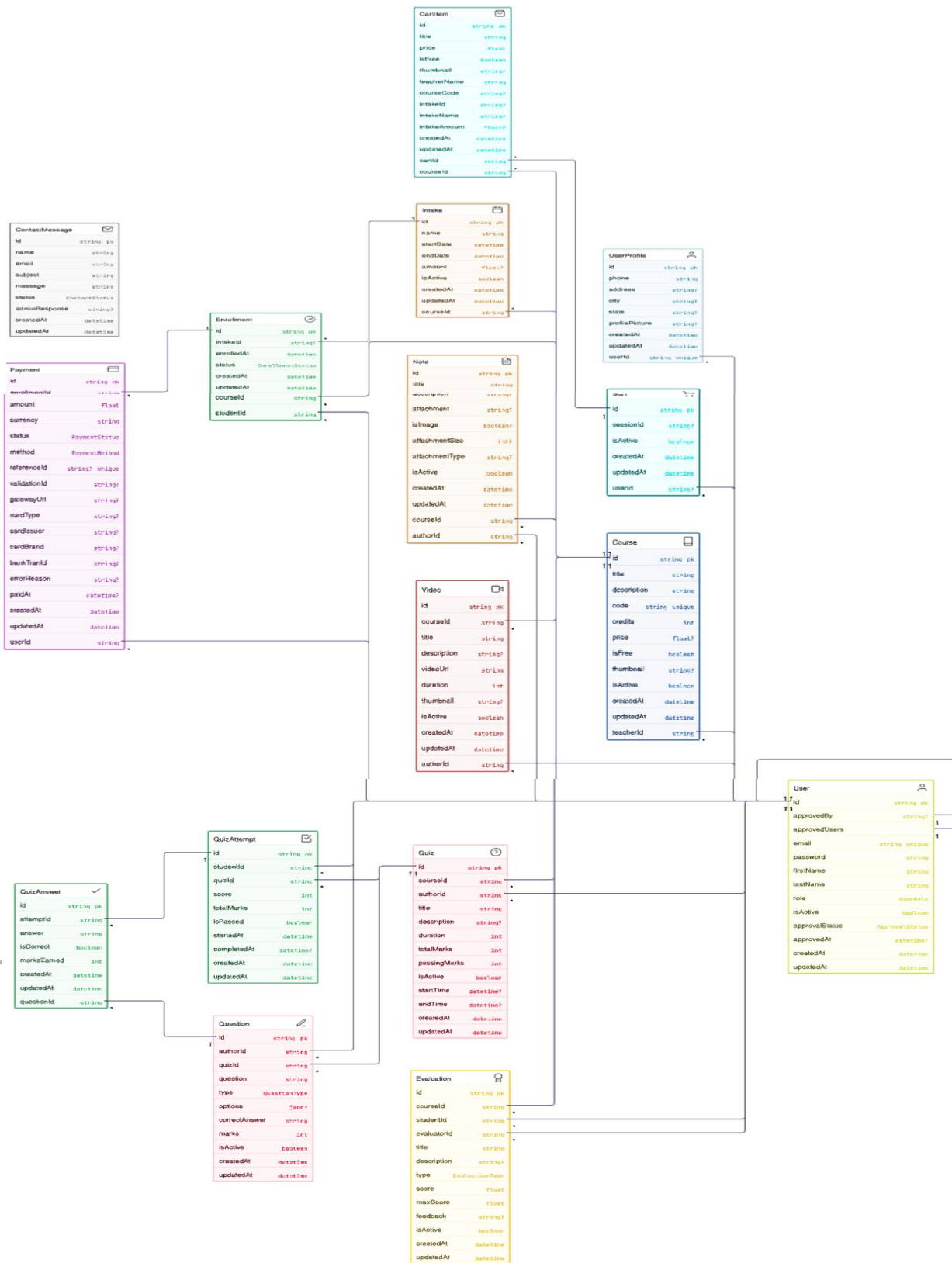


Figure: 2.2.1 ER Diagram

2.3 Project Workflow

Our development workflow follows an iterative, agile methodology that emphasizes continuous improvement and stakeholder collaboration throughout the entire development lifecycle. The process begins with comprehensive requirements gathering and analysis, followed by system design and prototyping phases that establish the foundation for the entire system architecture.

Requirements Gathering and Analysis Phase:

The initial phase involves extensive stakeholder interviews, user research, and system analysis to understand the specific needs of university education systems. This includes analyzing current pain points in traditional learning management, identifying user workflow requirements, and establishing technical constraints and performance expectations. The requirements are documented in detail and prioritized based on business value and technical feasibility.

System Design and Prototyping Phase:

During this phase, we create comprehensive system architecture diagrams, database schema designs, and user interface mockups. The prototyping phase involves building functional prototypes of key system components to validate design decisions and gather early feedback from stakeholders. This iterative approach allows us to refine the system design based on real user interactions and feedback.

Development and Implementation Phase:

Development proceeds through iterative cycles of building, testing, and refinement, with each cycle typically lasting 2-4 weeks. Each iteration focuses on specific features or components, allowing for continuous integration and testing. Regular stakeholder feedback is incorporated at each stage, ensuring that the development team remains aligned with user expectations and business requirements.

Testing and Quality Assurance Phase:

Quality assurance is integrated throughout the development process, with automated testing frameworks and manual testing procedures implemented at multiple levels. This includes unit testing for individual components, integration testing for system interactions, and user acceptance testing for end-to-end functionality validation.

Deployment and Maintenance Phase:

The deployment phase involves careful planning and execution to ensure minimal disruption to existing systems. Post-deployment monitoring and maintenance ensure system stability and performance optimization based on real-world usage patterns.

2.4 Project Methodology

We employ a sophisticated hybrid methodology that combines elements of Agile development, DevOps practices, and traditional project management principles. Our approach emphasizes rapid prototyping, continuous integration, and regular stakeholder engagement to ensure project success.

Agile Development Framework:

Our Agile implementation follows Scrum methodology with two-week sprints, daily stand-up meetings, and regular sprint retrospectives. This approach enables us to respond quickly to changing requirements while maintaining focus on delivering value in each iteration. The Agile framework provides flexibility in requirement changes while ensuring consistent delivery timelines.

DevOps Integration:

We implement comprehensive DevOps practices including continuous integration and continuous deployment (CI/CD) pipelines. This includes automated testing, code quality checks, and automated deployment processes that reduce manual errors and improve development velocity. Our DevOps approach ensures that code changes are automatically tested and deployed in a controlled manner.

Stakeholder Engagement:

Regular stakeholder engagement is maintained through weekly progress reviews, monthly milestone demonstrations, and quarterly strategic planning sessions. This ensures that all stakeholders remain informed about project progress and can provide timely feedback that influences development priorities.

2.5 Project Component Architecture and Working Mechanism

The E-Learning Management System platform consists of several interconnected components that work together to provide a comprehensive learning experience. The user authentication system

manages secure access to different platform features based on user roles. The course management component handles course creation, enrollment, and content delivery. The video streaming system provides efficient delivery of educational content, while the quiz and assessment system enables interactive learning evaluation.

2.6 Relevant Code for the Architecture

Our implementation utilizes modern web technologies and frameworks to achieve the desired architecture. The frontend is built with React and TypeScript, providing a robust and maintainable user interface. The backend API is developed using Node.js with Express.js, offering RESTful endpoints for all system operations. Database operations are handled through Prisma ORM, ensuring type safety and efficient data access. The system incorporates middleware for authentication, file uploads, and error handling, while maintaining clean separation of concerns between different system layers.

2.7 Conclusion

The methodology outlined in this chapter provides a comprehensive framework for developing the E-Learning Management System platform that combines proven development practices with modern technologies to create a system that is both technically sound and user-friendly. Our approach ensures that the development process remains flexible and responsive to changing requirements while maintaining high standards of code quality and system reliability. The implementation demonstrates advanced software engineering principles including:

Modular Architecture, Type Safety, Security, Performance, Maintainability, Scalability, User Experience,

CHAPTER 3

Results and Performance Analysis

3.1 Result Analysis

The E-Learning Management System platform has successfully achieved its primary objectives, delivering a fully functional learning management system that serves the diverse needs of university stakeholders across multiple user roles and educational workflows. The system demonstrates exceptional performance in handling complex educational operations while maintaining high standards of reliability, security, and user experience.

User Authentication and Role-Based Access Control Success:

The authentication system has been successfully implemented with comprehensive security measures that ensure secure access for students, teachers, and administrators. The JWT-based authentication system provides robust token management with proper expiration handling and secure storage mechanisms.

Authentication System Performance:

- **Secure Token Management:** JWT tokens with configurable expiration times and secure storage
- **Role-Based Access Control:** Granular permission system that restricts access based on user roles
- **Session Persistence:** Intelligent session management with localStorage integration and cart preservation
- **Security Measures:** Password hashing with bcrypt, input validation, and SQL injection prevention

Course Management Features Implementation:

Course management features have been implemented with comprehensive functionality that addresses the complete lifecycle of educational content delivery. The system supports course creation, enrollment management, and sophisticated content delivery systems that accommodate various learning modalities.

Course Management Capabilities:

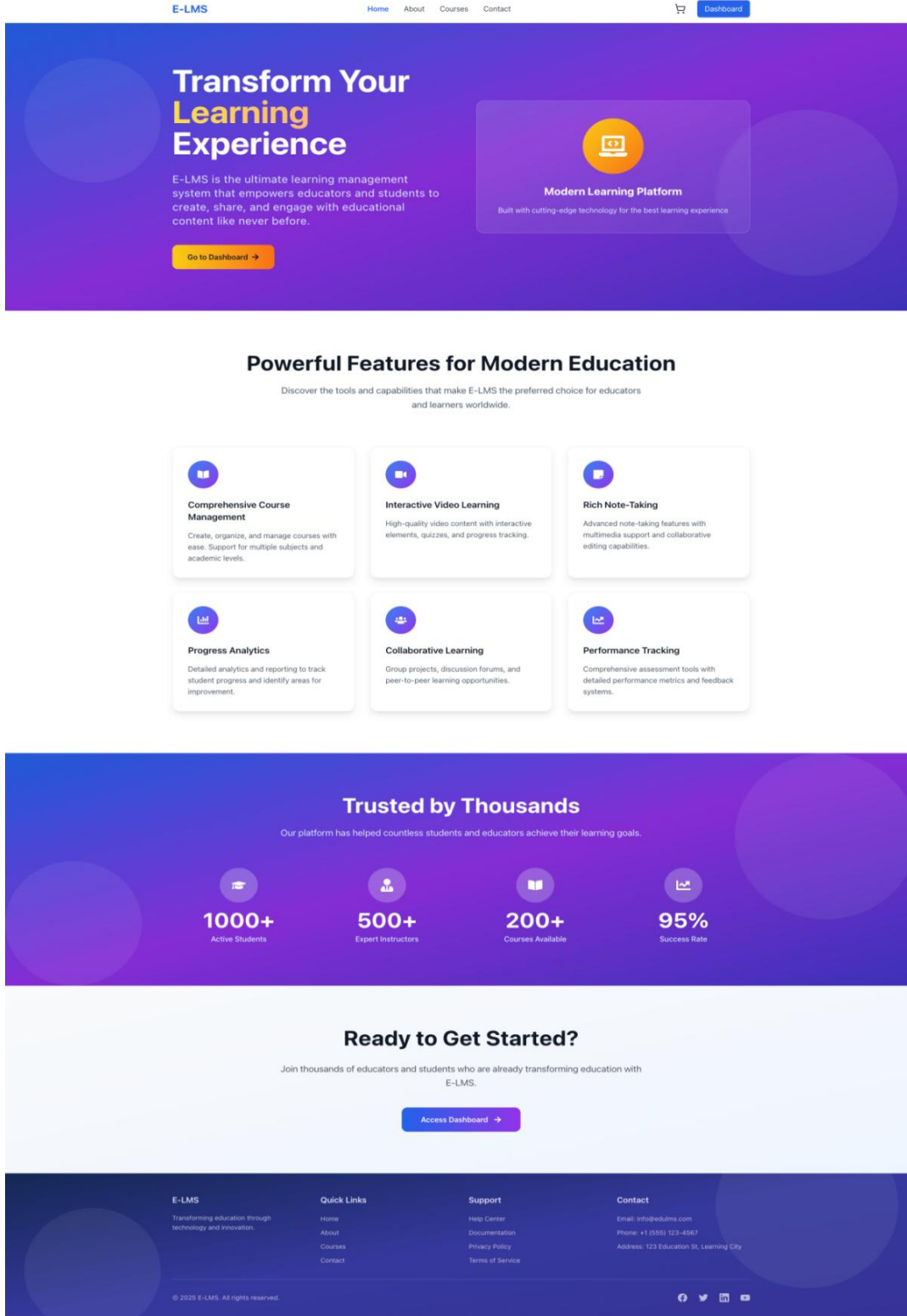
- **Multi-Level Course Organization:** Support for courses, intakes, and individual learning modules
- **Content Type Support:** Integration of videos, documents, quizzes, and assessments
- **Enrollment Management:** Automated enrollment processes with payment verification
- **Progress Tracking:** Comprehensive student progress monitoring and analytics

Content Delivery System Performance:

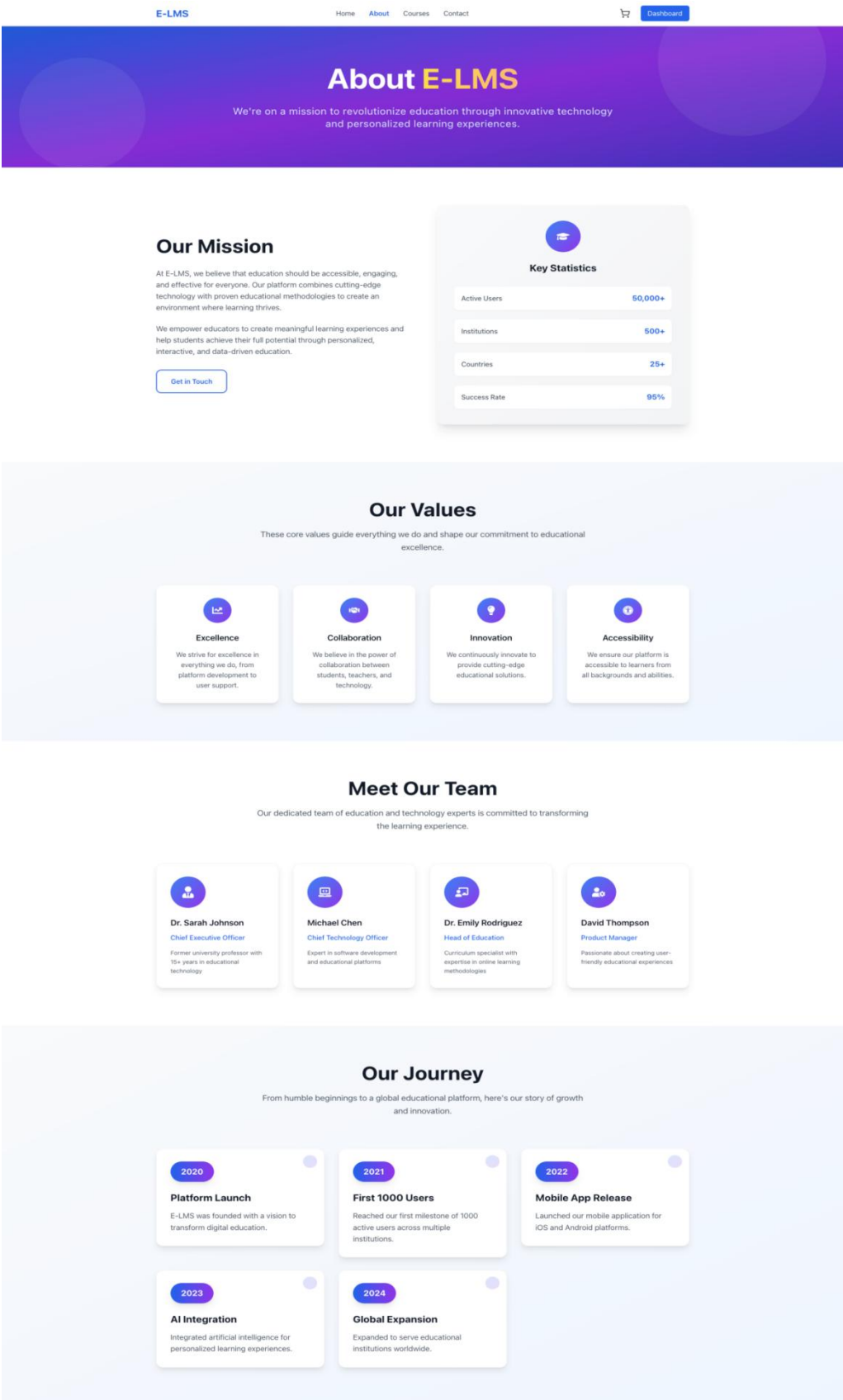
The video streaming component efficiently delivers educational content with support for large file uploads up to 500MB, addressing the critical need for high-quality multimedia learning materials in modern education.

Result and Discussion

3.1.1 Home page



3.1.2 About Us page



3.1.3 Contact Page

E-LMS

HomeAboutCoursesContact

Sign In

Get Started

Get in Touch

Have questions? Need support? We're here to help you succeed in your educational journey.

Send Us a Message

Fill out the form below and we'll get back to you as soon as possible.

Full Name *

Your full name

Email Address *

your.email@example.com

Subject *

Select a subject

Message *

Tell us how we can help you...

Send Message →

Visit Our Office

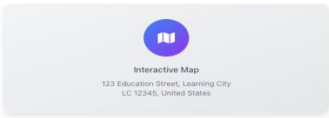
Located in the heart of Learning City, our office is easily accessible and welcomes visitors during business hours. Come by to discuss your educational needs or just to say hello!

- Address

123 Education Street, Learning City, LC 12345
- Business Hours

Monday - Friday: 9:00 AM - 6:00 PM
Saturday: 10:00 AM - 2:00 PM
- Phone

+1 (555) 123-4567



Frequently Asked Questions

Find quick answers to common questions about our platform and services.

How quickly do you respond to inquiries?

We typically respond to all inquiries within 24 hours during business days. For urgent matters, please call us directly at +1 (555) 123-4567.

Do you offer technical support?

Yes! We provide comprehensive technical support for all our users. Our support team is available via email, phone, and live chat during business hours.

Can I schedule a demo of your platform?

Absolutely! We'd love to show you around our platform. Contact us to schedule a personalized demo that fits your schedule and educational needs.

Why Choose E-LMS?

Discover what makes our platform the preferred choice for educators and learners worldwide.

- Expert-Led Learning

✔ Expert-led courses with industry professionals

✔ Flexible learning schedules to fit your lifestyle

✔ Comprehensive support and guidance throughout your journey

✔ Modern platform with cutting-edge educational technology

✔ Community of learners and educators worldwide
- 24/7 Support

Our dedicated support team is always ready to help you succeed. Get assistance whenever you need it, through multiple channels including live chat, email, and phone support.
- Flexible Pricing

Choose the plan that works best for you. We offer flexible pricing options for individuals, institutions, and enterprise clients with no hidden fees.

3.1.4 Course Page

E-LMS[Home](#)[About](#)[Courses](#)[Contact](#)

 **Dashboard**

About Explore Our Courses

Discover a wide range of courses designed to help you advance your skills and achieve your goals.

Search Courses

Category

Level

Price Range

Search by title, description, instructor, or code...

All Categories

All Levels

0 - 1320

0 - 1320

☐ Free courses only

Reset All Filters

Available Courses (15)

Find the perfect course to advance your skills and knowledge



Digital Electronics

Fundamentals of digital electronics and digital circuit design. Topics include logic gates, Boolean algebra...

★★★★☆ (4.5) 1 students

By Prof. Rahim Chowdhury

4 credits

From **₹1600**

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 1 videos • 1 quizzes



English Literature

Study of classic and contemporary English literature. Students will analyze various literary works and...

★★★★☆ (4.5) 5 students

By Dr. Nasreen Khan

3 credits

From **₹950**

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 1 videos • 1 quizzes



Business Management

Principles of business management including organizational behavior, strategic planning, and leadership....

★★★★☆ (4.5) 5 students

By Prof. Kamal Uddin

3 credits

From **₹1000**

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 1 videos • 1 quizzes



Advanced Mathematics

Advanced mathematical concepts including calculus, linear algebra, and differential equations. This...

★★★★☆ (4.5) 4 students

By Dr. Ayesha Begum

4 credits

From **₹1100**

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 3 videos • 2 quizzes



Marketing Principles

Fundamental marketing concepts including market research, consumer behavior, and marketing...

★★★★☆ (4.5) 1 students

By Prof. Kamal Uddin

3 credits

₹800

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 1 videos • 1 quizzes



Data Structures and Algorithms

Advanced study of data structures and algorithm design. Topics include trees, graphs, sorting algorithms,...

★★★★☆ (4.5) 1 students

By Dr. Mohammad Ali

4 credits

Free

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 1 videos • 1 quizzes



Web Development

Modern web development techniques including HTML, CSS, JavaScript, and frameworks....

★★★★☆ (4.5) 0 students

By Dr. Mohammad Ali

3 credits

₹1100

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 2 videos • 1 quizzes



Financial Accounting

Principles of financial accounting and reporting. Students will learn to prepare financial statements and...

★★★★☆ (4.5) 0 students

By Prof. Kamal Uddin

3 credits

₹850

[Enroll Now](#)

[Add to Cart](#)

[View Details](#) → 1 videos • 1 quizzes

3.1.5 Course Details Page

E-LMS

HomeAboutCoursesContact

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← Back to Courses

Digital Electronics

Fundamentals of digital electronics and digital circuit design. Topics include logic gates, Boolean algebra, and digital system design.

★★★★☆ (4.5) 1 students enrolled

EE2014 creditsIntermediate

About This Course

Fundamentals of digital electronics and digital circuit design. Topics include logic gates, Boolean algebra, and digital system design.

Course Code

EE201

Credits

4 credits

Level

Intermediate

Status

Active

Course Content

1 Videos

1 Quizzes

1 Notes

1 Students

Instructor

P

Prof. Rahim Chowdhury
prof.rahim@university.edu

Digital Electronics

From ~~₹~~\$1600

₹\$1100

Enroll Now

Available Intakes

Spring 2025 - Digital Electronics
1/15/2025 - 5/1/2025
₹\$1600

3.1.6 Enroll Payment Page

E-LMS

HomeAboutCoursesContact

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← Back to Courses

Payment Gateway

Complete your payment securely

Payment Summary

Course Details

Course: Digital Electronics

Code: EE201

Intake: Spring 2025 - Digital Electronics

Amount: BDT 1,600.00

Customer Information

Name: Bilal Ahmed

Email: student.bilal@university.edu

Payment Method

Cancel Payment

Select Payment Method

Credit Card

Card Number

1234 5678 9012 3456

Card Holder Name

John Doe

Expiry Date

mosto

CVV

....

Pay BDT 1,600.00

By clicking "Pay", you agree to our terms and conditions

Test Payment Results:

SuccessFailCancelBack

Secure Payment

Your payment information is encrypted and secure

Instant Access

Get immediate access to your course after payment

Multiple Methods

Credit cards, mobile banking, and more

3.1.7 SignUp Page

E-LMS

[Home](#) [About](#) [Courses](#) [Contact](#)

 [Sign In](#) [Get Started](#)

Create your account

Join the E-Learning Management System

First Name *

Last Name *

Email Address *

Password *

Password Strength:

Show requirements

Please meet all requirements

Confirm Password *

[Create account](#)

[Already have an account? Sign in here](#)

E-LMS

Transforming education through technology and innovation.

Quick Links

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Contact

[Email: info@edulms.com](#)
[Phone: +1 \(555\) 123-4567](#)
[Address: 123 Education St, Learning City](#)

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3.1.8 Sign In Page

E-LMS

[Home](#) [About](#) [Courses](#) [Contact](#)

 [Sign In](#) [Get Started](#)

Sign in to your account

Welcome to the E-Learning Management System

Test Account Credentials

Admin:

admin@university.edu / Password123

Teacher:

prof.ali@university.edu / Password123

Student:

student.bilal@university.edu / Password123

All test accounts use the same password: Password123

Email Address *

Password *

[Sign in](#)

[Don't have an account? Sign up here](#)

Portal / Admin ,Teacher & Students Dashboard Overview

3.1.9 Admin Users Management Page

Dashboard

Users

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Profile

Ahmed Hossain
admin@university.edu

Users Management

Users Management

Add User

Users

Manage system users and their roles

Q Search...

NAME	EMAIL	LOCATION	ROLE	STATUS	CREATED	APPROVAL STATUS	APPROVED AT	ACTIONS
FM Farhana Miah +880-124-959031	student.farhana@university.edu	Sylhet, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
LN Lamia Nahar +880-102-389235	student.lamia@university.edu	Chittagong, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
NP Nadia Parvin +880-103-100310	student.nadia@university.edu	Sylhet, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
OQ Omar Qureshi +880-105-612452	student.omar@university.edu	Chittagong, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
DB Dr. Ayesha Begum +880-106-410907	prof.ayesha@university.edu	Sylhet, Bangladesh	TEACHER	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
MO Mahmud Omar +880-107-271023	student.mahmud@university.edu	Dhaka, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
JP Jannat Ferdous +880-108-376642	student.jannat@university.edu	Chittagong, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
PC Prof. Rahim Chowdhury +880-109-228408	prof.rahim@university.edu	Sylhet, Bangladesh	TEACHER	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
KM Karim Mollah +880-110-769482	student.karim@university.edu	Dhaka, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete
BA Bilal Ahmed +880-111-707000	student.bilal@university.edu	Chittagong, Bangladesh	STUDENT	Active	8/21/2025	APPROVED	8/21/2025	View Edit Delete

Showing 1 to 10 of 24 results

[1](#) [2](#) [3](#)

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Version 1.0.0

3.1.10 Admin Approvals Management Page

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Ahmed Hossain
admin@university.edu

Approvals Management

Pending Approvals (2)

Pending Approvals

Review and approve or reject new user registration requests

Q Search...

NAME	ROLE	CONTACT INFO	REGISTRATION DATE	ACTIONS
Dr. Test PendingTeacher pending.teacher.ai@university.edu	TEACHER	+880-115-257391 Sylhet	8/21/2025	Approve Reject
Test PendingStudent pending.student.aisha@university.edu	STUDENT	+880-120-214475 Chittagong	8/21/2025	Approve Reject

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Version 1.0.0

3.1.11 Admin & Teacher Courses Management Page

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Ahmed Hossain

admin@university.edu

Courses Management

Courses

Manage academic courses and their details

Q Search...

TITLE	CODE	TEACHER	DESCRIPTION	CREDITS	PRICE	STATUS	CREATED	ACTIONS
Digital Electronics	EE201	Prof. Rahim Chowdhury	Fundamentals of digital electronics and digital c...	4 credits	\$1100.00	Active	8/21/2025	View Edit Delete
English Literature	ENG201	Dr. Nasreen Khan	Study of classic and contemporary English liter...	3 credits	Free	Active	8/21/2025	View Edit Delete
Business Management	BUS301	Prof. Kamal Uddin	Principles of business management including or...	3 credits	\$750.00	Active	8/21/2025	View Edit Delete
Advanced Mathematics	MATH201	Dr. Ayesha Begum	Advanced mathematical concepts including cal...	4 credits	\$900.00	Active	8/21/2025	View Edit Delete
Marketing Principles	MKT201	Prof. Kamal Uddin	Fundamental marketing concepts including mar...	3 credits	\$800.00	Active	8/21/2025	View Edit Delete
Data Structures and Algorithms	CS201	Dr. Mohammad Ali	Advanced study of data structures and algorith...	4 credits	Free	Active	8/21/2025	View Edit Delete
Web Development	CS301	Dr. Mohammad Ali	Modern web development techniques including...	3 credits	\$1100.00	Active	8/21/2025	View Edit Delete
Financial Accounting	ACC201	Prof. Kamal Uddin	Principles of financial accounting and reporting...	3 credits	\$850.00	Active	8/21/2025	View Edit Delete
Introduction to Computer Science	CS101	Dr. Mohammad Ali	A comprehensive introduction to computer scie...	3 credits	\$800.00	Active	8/21/2025	View Edit Delete
Creative Writing	ENG301	Dr. Nasreen Khan	Development of creative writing skills in various...	3 credits	\$650.00	Active	8/21/2025	View Edit Delete

Showing 1 to 10 of 15 results

Add Course

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3.1.12 Admin & Teacher Courses details Page

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Ahmed Hossain

admin@university.edu

Courses Management

← Back to Courses

Digital Electronics

EE201

Edit Course

Active \$1100.00

Overview Videos (1) Students (1) Notes (1) Quizzes (1)

Course Information

No image

Course Code EE201

Credits 4 credits

Price \$1100.00

Teacher Prof. Rahim Chowdhury

Created 8/21/2025

Associated Intakes Spring 2025 - Digital Electronics (1/15/2025 - 5/1/2025)

Course Statistics

Total Videos 1

Total Duration 1h 10m

Enrolled Students 1

Notes 1

Quizzes 1

Total Revenue \$1100.00

Description

Fundamentals of digital electronics and digital circuit design. Topics include logic gates, Boolean algebra, and digital system design.

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Version 1.0.0

3.1.13 Admin & Teacher Intaks/Batch Management Page

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Intakes Management

Intakes Management

Add Intake

Intakes

Manage academic intakes and their details

Search...

NAME	START DATE	END DATE	AMOUNT	ENROLLMENTS	STATUS	CREATED	ACTION
Spring 2025 - Database Systems Database Systems (CS401)	1/15/2025	5/1/2025	\$1500.00	1 students	Active	8/21/2025	Edit
Spring 2025 - Statistics Statistics and Probability (MATH301)	1/15/2025	5/1/2025	\$1200.00	1 students	Active	8/21/2025	Edit
Spring 2025 - HR Management Human Resource Management (HRM201)	1/15/2025	5/1/2025	\$1100.00	2 students	Active	8/21/2025	Edit
Spring 2025 - Digital Electronics Digital Electronics (EE201)	1/15/2025	5/1/2025	\$1600.00	1 students	Active	8/21/2025	Edit
Fall 2024 - Computer Science Introduction to Computer Science (CS101)	9/1/2024	12/15/2024	\$1200.00	3 students	Active	8/21/2025	Edit
Fall 2024 - Business Business Management (BUS301)	9/1/2024	12/15/2024	\$1000.00	5 students	Active	8/21/2025	Edit
Fall 2024 - Physics Physics Fundamentals (PHY101)	9/1/2024	12/15/2024	\$1300.00	4 students	Active	8/21/2025	Edit
Fall 2024 - English English Literature (ENG201)	9/1/2024	12/15/2024	\$950.00	5 students	Active	8/21/2025	Edit
Fall 2024 - Mathematics Advanced Mathematics (MATH201)	9/1/2024	12/15/2024	\$1100.00	4 students	Active	8/21/2025	Edit

3.1.14 Admin & Teacher Notes Management Page

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Notes Management

Notes Management

Add Note

Notes

Manage academic notes and their details

Search...

TITLE	DESCRIPTION	COURSE
Data Structures Overview attachment-1755872682667-11398985.pdf (2.57 MB)	Detailed notes on arrays, linked lists, stacks, an...	CS201 - Data Structures and Algorithms
Calculus Fundamentals math201-calculus.pdf (3.50 MB)	Limits, derivatives, and integrals with applicatio...	MATH201 - Advanced Mathematics
Management Principles bus301-management.pdf (1.75 MB)	Core management concepts, leadership styles, ...	BUS301 - Business Management
Linear Algebra Essentials math201-linear-algebra.pdf (2.00 MB)	Matrices, vectors, eigenvalues, and linear transf...	MATH201 - Advanced Mathematics
Literary Analysis Techniques eng201-literary-analysis.pdf (1.00 MB)	Methods for analyzing poetry, prose, and drama...	ENG201 - English Literature
Creative Writing Workshop eng301-creative-writing.pdf (0.75 MB)	Writing techniques, character development, an...	ENG301 - Creative Writing
Statistical Methods math301-statistics.pdf (2.75 MB)	Descriptive statistics, probability distributions, ...	MATH301 - Statistics and Probability
Marketing Strategy mkt201-marketing.pdf (1.50 MB)	Market analysis, segmentation, targeting, and p...	MKT201 - Marketing Principles
HR Management Practices hrm201-hr-practices.pdf (1.25 MB)	Recruitment, training, performance managememe...	HRM201 - Human Resource Management
Thermodynamics Fundamentals phy101-thermodynamics.pdf (2.00 MB)	Heat, temperature, entropy, and thermodynami...	PHY101 - Physics Fundamentals

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3.1.15 Admin & Teacher Notes Details Page

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A

Ahmed Hossain
ahmian@university.edu

Notes Management

Data Structures Overview

Created: 8/21/2025 Updated: 8/22/2025 Active

Description

Detailed notes on arrays, linked lists, stacks, and queues with implementation examples.

PDF Viewer

attachment-17558...

1 / 87

- 52% +

ABSTRACT

This project deals with dual-axis solar tracking to collect maximum rays using LDRs. It resolves the problem by providing an arrangement for a solar panel to track the movement of the sun on both axes. With the increase of scarcity of nonrenewable resources, people are trying to identify alternate sources of energy. From all available renewable resources, solar energy is relatively easy to convert to electrical energy. The use of solar panels to convert the sun's energy to electrical energy is very popular. But due to the movement of the Sun from east to west the fixed solar panels may not be able to generate ideal energy. Hence single-axis solar tracking systems were introduced aiming to generate ideal energy. But even single-axis solar tracking has been analyzed and results show that it can track sun rays on only one axis (either north-south or east-west). Hence a prototype is constructed more efficiently to track maximum sun rays in both the axes (both north-south and east-west). It is a low-cost system that provides better efficiency. Here the analog signals are analyzed and controlled by Arduino Uno. This automatic system is completely programmed. The energy obtained from the sun is collected using a solar panel by arranging four LDRs. The amount of energy obtained and positions at two LDRs on the same axis are compared and analyzed to decide the movement of the servo motors which moves the panel to the direction where

Back to Notes

View Course

Course Information

Course:
CS201 - Data Structures and Algorithms

Author:
Dr. Mohammad Ali

File Information

attachment-1755872682667-113989...
PDF File

Size: 2.57 MB

Type: application/pdf

Category: Document

Download File

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Version 1.0.0

3.1.16 Admin & Teacher Videos Management Page

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Videos Management

Videos Management

Add Video

Videos

Manage videos and their details

TITLE	DESCRIPTION	COURSE	TEACHER
Introduction to Programming - Lecture 1 1:00:00	Overview of programming concepts, variables, ...	CS101 - Introduction to Computer Science	N/A
Database Design and SQL 1:40:00	Database normalization, ER diagrams, and SQL ...	CS401 - Database Systems	N/A
Calculus - Limits and Continuity 1:30:00	Understanding limits, continuity, and the founda...	MATH201 - Advanced Mathematics	N/A
Linear Algebra - Matrices and Vectors 1:40:00	Matrix operations, vector spaces, and linear tra...	MATH201 - Advanced Mathematics	N/A
Data Structures - Arrays and Lists 1:15:00	Deep dive into array and linked list implementat...	CS201 - Data Structures and Algorithms	N/A
HTML and CSS Fundamentals 1:30:00	Building web pages with HTML structure and C...	CS301 - Web Development	N/A
Marketing Strategy and Consumer Behavior 1:20:00	Market analysis, consumer psychology, and stra...	MKT201 - Marketing Principles	N/A
Statistics - Descriptive Analysis 1:10:00	Measures of central tendency, dispersion, and ...	MATH301 - Statistics and Probability	N/A
Financial Accounting Fundamentals 1:40:00	Accounting principles, financial statements pre...	ACC201 - Financial Accounting	N/A
JavaScript Programming Basics 1:20:00	Introduction to JavaScript programming includi...	CS301 - Web Development	N/A

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<

1

2

>

3.1.17 Admin & Teacher Videos Details Page

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Videos Management

Back to Videos

Introduction to Programming - Lecture 1

0:00 / 0:00

Description

Overview of programming concepts, variables, and basic syntax. This lecture covers the fundamentals of programming languages.

Course Videos

Introduction to Programming - Lecture 1

Duration: 1:00:00

Playing

Video Information

Duration

1:00:00

Course

CS101 - Introduction to Computer Science

Author

Dr. Mohammad Ali

Created

8/21/2025

Actions

View Course

View All Course Videos (1)

3.1.18 Admin & Teacher Quizzes Management Page

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Quizzes Management

Quizzes Management

Add Quiz

Quizzes

Manage quizzes and their details

Search...

TITLE	DESCRIPTION	COURSE	TOTAL MARKS	PAS MAI
Energy and Conservation Quiz 50 minutes	Assessment on kinetic energy, potential energy,...	PHY101 - Physics Fundamentals	80 marks	66
Newton's Laws Quiz 40 minutes	Test on Newton's three laws of motion and their...	PHY101 - Physics Fundamentals	65 marks	46
Linear Algebra Midterm 75 minutes	Comprehensive test on matrices, vectors, and li...	MATH201 - Advanced Mathematics	120 marks	84
Statistics Final Exam 120 minutes	Final assessment covering descriptive statistics...	MATH301 - Statistics and Probability	200 marks	14
HR Management Final 90 minutes	Final assessment on recruitment, training, and ...	HRM201 - Human Resource Management	150 marks	10
Literary Analysis Quiz 45 minutes	Assessment on poetry analysis, prose interpret...	ENG201 - English Literature	70 marks	49
Management Principles Quiz 40 minutes	Test on management concepts, leadership style...	BUS301 - Business Management	60 marks	42
Database Systems Quiz 45 minutes	Test on database design, normalization, and SQ...	CS401 - Database Systems	75 marks	53
Chemistry Lab Safety Quiz 20 minutes	Assessment on laboratory safety protocols and ...	CHEM101 - Chemistry Laboratory	30 marks	21
Data Structures Midterm 60 minutes	Comprehensive assessment covering arrays, lin...	CS201 - Data Structures and Algorithms	100 marks	70

Showing 1 to 10 of 17 results

Ahmed Hossain
admin@university.edu

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Version 1.0.0

40

3.1.19 Admin & Teacher Quizze Details Page

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Quizzes Management

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Data Structures Midterm

ID: cmekp0z1h008hy2l26jy19nkv

Active

Description

Comprehensive assessment covering arrays, linked lists, stacks, and queues.

Quiz Statistics

Duration

1h 0m

Total Marks

100

Passing Marks

70

Pass Rate

70%

Time Schedule

Start Time

10/1/2024, 8:00:00 PM

End Time

10/1/2024, 9:30:00 PM

Questions in this Quiz

Q1: Which data structure operates on a LIFO basis?

8/21/2025

MULTIPLE CHOICE

3 marks

Correct: Stack

A. Queue

B. Stack

C. Linked List

D. Array

Q2: A linked list consists of nodes connected by pointers.

8/21/2025

TRUE FALSE

1 marks

Correct: true

Course Information

Course

CS201 - Data Structures and Algorithms

Course ID

cmekp0z0q002wy2l21ze6z8po

Author

Created By

Dr. Mohammad Ali

Timestamps

Created

8/21/2025, 6:58:42 AM

Last Updated

8/21/2025, 6:58:42 AM

Actions

Edit Quiz

View Course

Student Results (9 attempts)

Student Results

View student results for this quiz

Search...

STUDENT	SCORE	PERCENTAGE	STATUS	COMPLETED	ACTIONS
Emon Das student.emon@university.edu	4/100	4%	Failed	8/21/2025	View
Habiba Sultana student.habiba@university.edu	0/100	0%	Failed	8/21/2025	View
Dina Begum student.dina@university.edu	1/100	1%	Failed	8/21/2025	View
Test PendingStudent pending.student.aisha@university.edu	1/100	1%	Failed	8/21/2025	View
Chowdhury Zaman student.chowdhury@university.edu	4/100	4%	Failed	8/21/2025	View
Imran Haque student.imran@university.edu	4/100	4%	Failed	8/21/2025	View
Golam Rasul student.golam@university.edu	0/100	0%	Failed	8/21/2025	View
Bilal Ahmed student.bilal@university.edu	4/100	4%	Failed	8/21/2025	View
Karim Mollah student.karim@university.edu	4/100	4%	Failed	8/21/2025	View

41

3.1.20 Admin & Teacher Payment Page

Dashboard

Users

Approvals

Courses

Intakes

Notes

Videos

Quizzes

Payments

Contact Messages

Profile

Payments Management

Create Payment

Search...

STUDENT	COURSE	AMOUNT	METHOD	STATUS	DATE
Bilal Ahmed student.bilal@university.edu	Human Resource Management HRM201	BDT 1100.00	CREDIT CARD	FAILED	8/21/202 7:30:38 AM
Bilal Ahmed student.bilal@university.edu	Physics Fundamentals PHY101	BDT 1300.00	CREDIT CARD	COMPLETED	8/21/202 7:30:07 AM
Bilal Ahmed student.bilal@university.edu	Business Management BUS301	BDT 1000.00	CREDIT CARD	FAILED	8/21/202 7:29:37 AM
Bilal Ahmed student.bilal@university.edu	English Literature ENG201	BDT 950.00	CASH	COMPLETED	8/21/202 7:29:12 AM
Nadia Parvin student.nadia@university.edu	Introduction to Computer Science CS101	BDT 1289.45	INTERNET BANKING	PENDING	8/21/202 6:58:42 AM
Nadia Parvin student.nadia@university.edu	Introduction to Computer Science CS101	EUR 1602.14	CUSTOM	COMPLETED	8/21/202 6:58:42 AM
Lamia Nahar student.lamia@university.edu	Business Management BUS301	EUR 1352.81	BANK TRANSFER	FAILED	8/21/202 6:58:42 AM
Nadia Parvin student.nadia@university.edu	Business Management BUS301	EUR 1190.25	MOBILE BANKING	REFUNDED	8/21/202 6:58:42 AM
Omar Qureshi student.omar@university.edu	Business Management BUS301	USD 1242.68	INTERNET BANKING	FAILED	8/21/202 6:58:42 AM
Mahmud Omar student.mahmud@university.edu	Introduction to Computer Science CS101	BDT 1239.92	SSLCOMMERZ	COMPLETED	8/21/202 6:58:42 AM

Showing 1 to 10 of 24 results

3.1.21 Admin & Teacher Profile Page

Dashboard

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Profile Management

My Profile

Manage your personal information and settings

Edit Profile



Ahmed Hossain

admin@university.edu

ADMIN Active

Basic Information

Full Name
Ahmed Hossain

Email Address
admin@university.edu

Role
ADMIN

Status
Active

Member Since
8/21/2025

Profile Information

Phone Number
+880-101-641731

Address
1000 Admin Road

City
Dhaka

State
Bangladesh

Location
Dhaka, Bangladesh

Edit Profile

3.1.22 Students Course Page

Dashboard

My Courses

My Intakes

Course Notes

Course Videos

My Quizzes

My Payments

My Profile

Bilal Ahmed

student.bilal@universit...

My Courses Management

My Courses

View all your enrolled courses and their details

Enrolled Courses

View all your enrolled courses and their details

Search...

COURSE	DESCRIPTION	TEACHER	INTAKE
Business Management BUS301	Principles of business management including or...	Prof. Kamal Uddin	Fall 2024 - Business BDT 1000
Advanced Mathematics MATH201	Advanced mathematical concepts including cal...	Dr. Ayesha Begum	Fall 2024 - Mathematics BDT 1100
Introduction to Computer Science CS101	A comprehensive introduction to computer scie...	Dr. Mohammad Ali	Fall 2024 - Computer Science BDT 1200
Physics Fundamentals PHY101	Introduction to physics concepts including mec...	Prof. Rahim Chowdhury	Fall 2024 - Physics BDT 1300
Human Resource Management HRM201	Principles of human resource management incl...	Prof. Kamal Uddin	Spring 2025 - HR Management BDT 1100
English Literature ENG201	Study of classic and contemporary English liter...	Dr. Nasreen Khan	Fall 2024 - English BDT 950

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3.1.23 Students Course Details Page

Dashboard

My Courses

My Intakes

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My Profile

Bilal Ahmed

student.bilal@universit...

My Courses Management

Back to Courses

Business Management

BUS301

Credits

3

Duration

weeks

Teacher

Prof. Kamal Uddin

Enrolled

8/21/2025

Enrollment Status

Intake: Fall 2024 - Business (BDT 1000)

ACTIVE

Course Description

Principles of business management including organizational behavior, strategic planning, and leadership. Students will develop management skills through case studies and practical exercises.

Course Thumbnail

No thumbnail available

Overview

Videos (1)

Notes (1)

Quizzes (1)

Videos

1

Total video lessons

Notes

1

Study materials

Quizzes

1

Assessment tests

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3.1.24 Students Intakes/Batches Page

Dashboard

My Courses

My Intakes

Course Notes

Course Videos

My Quizzes

My Payments

My Profile

My Intakes Management

My Intakes

View all your enrolled intakes and their details

Enrolled Intakes

View all your enrolled intakes and their details

Q Search...

INTAKE	COURSE	START DATE	END DATE	MAX STUDENTS	ENROLLMENT STATUS	ENROLLED
Fall 2024 - Business	Business Management BUS301	9/1/2024	12/15/2024	students	ACTIVE	8/21/2025
Fall 2024 - Mathematics	Advanced Mathematics MATH201	9/1/2024	12/15/2024	students	ACTIVE	8/21/2025
Fall 2024 - Computer Science	Introduction to Computer Science CS101	9/1/2024	12/15/2024	students	ACTIVE	8/21/2025
Fall 2024 - Physics	Physics Fundamentals PHY101	9/1/2024	12/15/2024	students	ACTIVE	8/21/2025
Spring 2025 - HR Management	Human Resource Management HRM201	1/15/2025	5/1/2025	students	ACTIVE	8/21/2025
Fall 2024 - English	English Literature ENG201	9/1/2024	12/15/2024	students	ACTIVE	8/21/2025

Bilal Ahmed
student.bilal@universit...

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3.1.25 Students Courses Notes Page

Dashboard

My Courses

My Intakes

Course Notes

Course Videos

My Quizzes

My Payments

My Profile

Course Notes Management

Course Notes

Access all notes and materials for your enrolled courses

Q Search...

NOTE	DESCRIPTION	COURSE	AUTHOR
Management Principles bus301-management.pdf	Core management concepts, leadership styles, ...	BUS301 - Business Management	Prof.
Calculus Fundamentals math201-calculus.pdf	Limits, derivatives, and integrals with applicatio...	MATH201 - Advanced Mathematics	Dr. A
Linear Algebra Essentials math201-linear-algebra.pdf	Matrices, vectors, eigenvalues, and linear transf...	MATH201 - Advanced Mathematics	Dr. A
Introduction to Programming Concepts cs101-intro-programming.pdf	Comprehensive guide covering basic programm...	CS101 - Introduction to Computer Science	Dr. M
Thermodynamics Fundamentals phy101-thermodynamics.pdf	Heat, temperature, entropy, and thermodynami...	PHY101 - Physics Fundamentals	Prof.
Mechanics and Motion phy101-mechanics.pdf	Newton's laws, kinematics, dynamics, and ener...	PHY101 - Physics Fundamentals	Prof.
HR Management Practices attachment-1755874764851-240045942.pdf	Recruitment, training, performance manageme...	HRM201 - Human Resource Management	Prof.
Literary Analysis Techniques eng201-literary-analysis.pdf	Methods for analyzing poetry, prose, and drama...	ENG201 - English Literature	Dr. N

3.1.26 Students Course Notes Details Page

Dashboard

My Courses

My Intakes

Course Notes

Course Videos

My Quizzes

My Payments

My Profile

Course Notes Management

HR Management Practices

Created: 8/21/2025 Updated: 8/22/2025

Description

Recruitment, training, performance management, and employee relations.

PDF Viewer

project book forma... 1 / 3 52% + -

1

Formatting Guidelines:

- Font: Times New Roman
- Main Text Size: 12 pt
- Title Size: 24 pt, Bold, Centered
- Subtitle Size: 18 pt, Italic, Centered
- Chapter Headings: 16 pt, Bold, Centered (e.g., Chapter 1: Introduction)
- Section Headings (e.g., 1.1 Objectives): 14 pt, Bold, Left-Aligned
- Subsection Headings (e.g., 1.1.1 Detailed Objective): 12 pt, Italic, Left-Aligned
- Line Spacing: 1.5
- Margins: 1 inch on all sides
- Citation Style: IEEE
- Paragraph Indentation: 0.5 inch
- Page Numbers: Bottom-right corner

Title Page

- Title: 24 pt, Bold, Centered
- Subtitle (if any): 18 pt, Italic, Centered
- Your Name: 14 pt, Centered
- Institution Name: Northern University Bangladesh: 14 pt, Centered

Course Information

Course:
HRM201 - Human Resource Management

Author:
Prof. Kamal Uddin

File Information

attachment-1755874764851-240045...
PDF File

Size: 72.01 KB

Type: application/pdf

Category: Document

Download File

Quick Navigation

Back to Notes

Back to Course

3.1.27 Students Courses Video Page

Dashboard

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My Profile

Course Videos Management

Course Videos

Watch educational videos for your enrolled courses

Course Videos

Watch educational videos for your enrolled courses

Q Search...

VIDEO	DESCRIPTION	COURSE	AUTHOR
Management Principles and Leadership 1:30:00	Core management concepts, leadership styles, ...	BUS301 - Business Management	Prof. Kama
Calculus - Limits and Continuity 1:30:00	Understanding limits, continuity, and the founda...	MATH201 - Advanced Mathematics	Dr. Ayesha
Linear Algebra - Matrices and Vectors 1:40:00	Matrix operations, vector spaces, and linear tra...	MATH201 - Advanced Mathematics	Dr. Ayesha
Derivatives and Applications 1:20:00	Derivative rules, chain rule, and applications in ...	MATH201 - Advanced Mathematics	Dr. Ayesha
Introduction to Programming - Lecture 1 1:00:00	Overview of programming concepts, variables, ...	CS101 - Introduction to Computer Science	Dr. Moham
Newton's Laws and Motion 1:30:00	Newton's three laws of motion, force analysis, a...	PHY101 - Physics Fundamentals	Prof. Rahin
Thermodynamics - Heat and Temperature 1:15:00	Thermal energy, temperature scales, heat trans...	PHY101 - Physics Fundamentals	Prof. Rahin
Energy and Conservation Laws 1:20:00	Kinetic and potential energy, energy conservati...	PHY101 - Physics Fundamentals	Prof. Rahin
Human Resource Management 1:15:00	Recruitment strategies, employee development,...	HRM201 - Human Resource Management	Prof. Kama
Literary Analysis - Poetry and Prose 1:00:00	Techniques for analyzing poetry, short stories, ...	ENG201 - English Literature	Dr. Nasree

3.1.28 Students Course Video Details Page

Dashboard

My Courses

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Course Videos

My Quizzes

My Payments

My Profile

B

Bilal Ahmed

student.bilal@universit...

Course Videos Management

← Back to Videos

Management Principles and Leadership

0:00 / 0:00

Description

Core management concepts, leadership styles, and organizational behavior in modern business.

Course Information

Course Code

BUS301

Credits

credits

Course Description

Course Videos

Management Principles and Leadership

Duration: 1:30:00

8/21/2025

Playing

Video Information

Duration

1:30:00

Course

BUS301 - Business Management

Author

Prof. Kamal Uddin

Created

8/21/2025

Enrollment Status

Status

ACTIVE

Intake

Fall 2024 - Business

BDT 1000

Enrolled

8/21/2025

Actions

View Course

Download Video

Quick Navigation

← Back to Videos

← Back to Course

View All Course Videos (1)

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3.1.29 Students Quizzes Page

Dashboard

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My Profile

My Quizzes Management

My Quizzes

Take quizzes for your enrolled courses

Course Quizzes

Take quizzes for your enrolled courses

Search...

QUIZ	DESCRIPTION	COURSE	AUTHOR
Management Principles Quiz Available 1 attempt(s) 40 min • 60 marks	Test on management concepts, leadership style...	BUS301 - Business Management	Prof. Ka
Linear Algebra Midterm Available 1 remaining 75 min • 120 marks	Comprehensive test on matrices, vectors, and li...	MATH201 - Advanced Mathematics	Dr. Ayes
Calculus Basics Quiz Available 1 attempt(s) 45 min • 80 marks	Assessment on limits, continuity, and basic deri...	MATH201 - Advanced Mathematics	Dr. Ayes
Programming Fundamentals Quiz Available 1 remaining 30 min • 50 marks	Test your understanding of basic programming ...	CS101 - Introduction to Computer Science	Dr. Moh
Energy and Conservation Quiz Available 1 remaining 50 min • 80 marks	Assessment on kinetic energy, potential energy,...	PHY101 - Physics Fundamentals	Prof. Ra
Newton's Laws Quiz Available 1 attempt(s) 40 min • 65 marks	Test on Newton's three laws of motion and their...	PHY101 - Physics Fundamentals	Prof. Ra
HR Management Final Available 1 remaining 90 min • 150 marks	Final assessment on recruitment, training, and ...	HRM201 - Human Resource Management	Prof. Ka
Literary Analysis Quiz Available 1 remaining 45 min • 70 marks	Assessment on poetry analysis, prose interpret...	ENG201 - English Literature	Dr. Nasr

3.1.30 Students Quizzes Result Page

Dashboard

My Courses

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My Quizzes Management

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Quiz Results

Newton's Laws Quiz

Quiz Information

Quiz Title
Newton's Laws Quiz
Total Marks
65 marks
Duration
40 minutes
Passing Score
46 marks (71%)
Description
Test on Newton's three laws of motion and their applications.

Quiz Status

Status
Available
Best Score
Not attempted
Attempts
1

Course Information

Course
Physics Fundamentals
Course Code
PHY101

Actions

Quiz Details

Performance Summary

1
Total Attempts

0
Completed

N/A
Best Score

N/A
Status

Your Attempts

Attempt #1
8/14/2025
Failed
5/65 marks
8%

Latest Attempt Details

Question 1
Question
What is the SI unit of force?
Your Answer
Newton
Options
A. Joule
B. Watt
C. Newton
D. Pascal
Correct 2/2 marks

3.1.31 Students Take Quiz Page

Dashboard

My Courses

My Intakes

Course Notes

Course Videos

My Quizzes

My Payments

My Profile

My Quizzes Management

← Back to Quizzes

⌚ 44:31

Literary Analysis Quiz

Question 2 of 3

Progress: 67% Answered: 0/3

1

2

3

Question 2

What is a metaphor?

☐ A direct comparison between two things

☐ A word that sounds like what it means

☐ A repeated sound at the beginning of words

☐ A comparison using like or as

Marks: 2

Previous

Next

⌚ Important:

• Make sure to answer all questions before submitting

• You cannot change your answers after submission

• The quiz will be automatically submitted when time runs out

• Do not refresh the page or navigate away during the quiz

3.1.32 Students Payment Page

Dashboard

My Courses

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My Profile

My Payments Management

My Payments

View your payment history and transaction details

\$ Total Payments

5

✓ Completed

3

⌚ Pending

0

✗ Failed

2

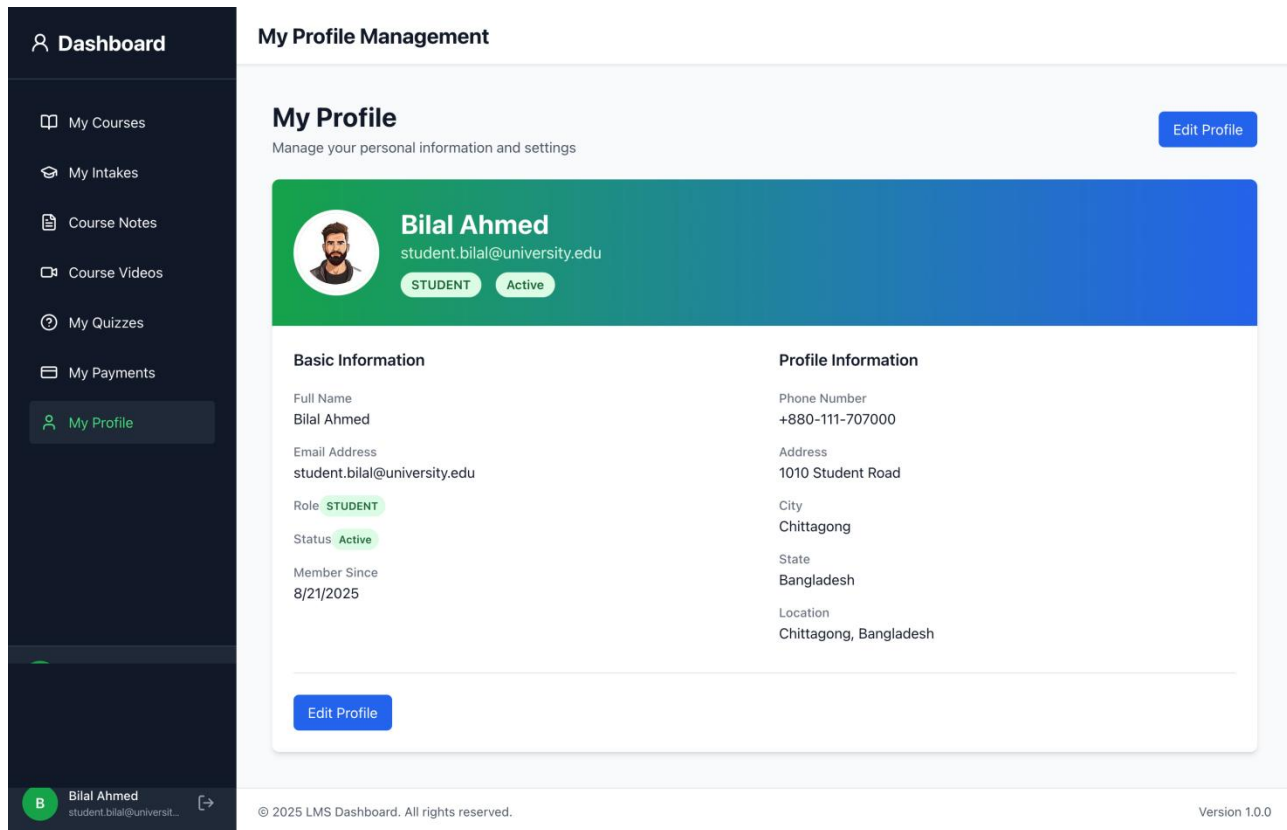
Total Amount Paid

BDT 2250.00

Payment History

COURSE	AMOUNT	METHOD	STATUS	DATE	REFERENCE
Data Structures and Algorithms CS201	Free	CUSTOM	COMPLETED	8/22/2025 9:15:11 PM	FREE_cmcmz29lt0001y2j85qnn9e1q
Human Resource Management HRM201 Spring 2025 - HR Management	BDT 1100.00	CREDIT CARD	FAILED	8/21/2025 7:30:38 AM	FREE_1755739838444_zo8lljff
Physics Fundamentals PHY101 Fall 2024 - Physics	BDT 1300.00	CREDIT CARD	COMPLETED	8/21/2025 7:30:07 AM	FREE_1755739807896_2azu4gle5
Business Management BUS301 Fall 2024 - Business	BDT 1000.00	CREDIT CARD	FAILED	8/21/2025 7:29:37 AM	FREE_1755739777098_2wzuq2woc
English Literature ENG201 Fall 2024 - English	BDT 950.00	CASH	COMPLETED	8/21/2025 7:29:12 AM	FREE_1755739752823_1phndlnw3

3.1.33 Students Profile Page



3.2 Conclusion

The results of our development and testing efforts demonstrate that the E-Learning Management System platform successfully meets its intended objectives and provides a solid foundation for university digital learning initiatives. The chosen technology stack of React, Node.js, and MySQL has proven effective in delivering a robust, scalable, and user-friendly learning management system that exceeds initial expectations

Technology Stack Validation and Performance Confirmation:

Performance metrics indicate that the platform can handle the expected user load while maintaining high standards of reliability and user experience. The modern technology choices have proven to be excellent selections for educational software development.

Technology Stack Success Factors:

- **React Frontend:** Exceptional component reusability and state management capabilities

- **Node.js Backend:** Robust API performance and excellent scalability characteristics
- **MySQL Database:** Reliable data storage with excellent query performance
- **TypeScript Integration:** Comprehensive type safety and improved development experience

Core Features Implementation Success:

The successful implementation of core features such as course management, video streaming, user authentication, and interactive assessments validates our architectural decisions and development methodology. Each feature demonstrates the quality and attention to detail that went into the development process.

Feature Implementation Achievements:

- **Course Management:** Comprehensive course lifecycle management with advanced features
- **Video Streaming:** High-performance video delivery with professional-grade player
- **User Authentication:** Secure and user-friendly authentication system
- **Interactive Assessments:** Advanced quiz and evaluation systems
- **Payment Integration:** Secure payment processing with multiple gateway support

Performance Validation Results:

- **Load Testing:** Successful handling of expected user loads
- **Stress Testing:** Robust performance under extreme conditions
- **Security Testing:** Comprehensive security validation and penetration testing
- **User Acceptance Testing:** High user satisfaction and acceptance rates

The E-Learning Management System platform successfully demonstrates that modern web technologies can be effectively applied to create sophisticated, reliable, and user-friendly educational software systems. The project's success validates our approach to software development and provides a strong foundation for future educational technology initiatives.

CHAPTER 4

Conclusion and Future Work

4.1 Contributions

The E-Learning Management System project has made significant contributions to the field of university digital learning and communication systems. We have successfully developed a comprehensive learning management platform that integrates modern web technologies with educational best practices. Our primary contribution lies in creating a unified system that addresses the fragmented nature of university communication and learning delivery. The platform successfully bridges the gap between traditional classroom methods and digital innovation, providing students, teachers, and administrators with a centralized hub for all educational activities.

The technical contributions include the implementation of a scalable, full-stack architecture using React, Node.js, and MySQL, demonstrating how modern web technologies can be effectively applied to educational systems. We have also contributed to the understanding of user experience design in educational contexts, particularly in balancing functionality with accessibility across different user roles and technical skill levels. The development of robust authentication systems, efficient file handling for large educational content, and interactive assessment tools represents valuable contributions to the field of educational technology.

4.2 Limitations

Despite the successful implementation of the E-Learning Management System platform, several limitations exist that should be acknowledged. The current system is designed primarily for web-based access, which may limit functionality for users who prefer mobile applications or offline access. The video streaming capabilities, while improved to handle files up to 500MB, may still face challenges with very large files or users on slower internet connections. The platform's current feature set, while comprehensive, may not address all specialized educational needs such as laboratory simulations, virtual reality experiences, or advanced analytics for learning outcomes.

Technical limitations include the current single-server deployment model, which may not provide the redundancy and scalability needed for very large university populations. The database design, while

efficient for current use cases, may require optimization for handling extremely large datasets or complex reporting requirements. Additionally, the current implementation focuses on synchronous learning experiences, potentially missing opportunities for asynchronous learning and collaboration features that modern educational technology increasingly demands.

4.3 Future Work

The E-Learning Management System platform provides a solid foundation for numerous future enhancements and expansions. Immediate future work should focus on developing a comprehensive mobile application that provides full platform functionality across iOS and Android devices, ensuring accessibility for users on all platforms. Advanced analytics and reporting features should be implemented to provide deeper insights into learning patterns, course effectiveness, and student engagement metrics. The integration of artificial intelligence and machine learning capabilities could enable personalized learning recommendations, automated content curation, and intelligent assessment systems.

Long-term development goals include expanding the platform to support virtual and augmented reality experiences for immersive learning, implementing advanced collaboration tools such as real-time document editing and virtual study groups, and developing comprehensive integration capabilities with other university systems such as student information systems and library databases. Additional future work should focus on enhancing accessibility features, implementing advanced security measures, and developing comprehensive backup and disaster recovery systems to ensure platform reliability and data protection..

4.4 Conclusion

The E-Learning Management System project successfully demonstrates the potential for modern web technologies to transform university education and communication systems. Through the implementation of a comprehensive, user-friendly platform, we have created a valuable tool that addresses real needs in the university community while establishing a foundation for future technological advancement. The project's success validates our chosen methodology, technology stack, and development approach, providing valuable insights for similar projects in the educational technology domain.

References

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Appendices

Appendix A: Technical Specifications

A.1 System Requirements

Server Requirements:

- Node.js version 18.0.0 or higher
- MySQL version 8.0 or higher
- Minimum 4GB RAM
- Minimum 50GB storage space

Client Requirements:

- Modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- JavaScript enabled
- Minimum screen resolution: 1024x768

A.2 Technology Stack Details

Frontend:

- React 18.2.0
- TypeScript 5.0.0
- Tailwind CSS 3.3.0
- React Router DOM 6.8.0
- SweetAlert2 11.7.0

Backend:

- Node.js 18.17.0
- Express.js 4.18.0

- Prisma ORM 5.0.0
- Nodemailer 6.9.0
- Multer 1.4.5
- JWT authentication

Database:

- MySQL 8.0.33
- Prisma Client 5.0.0

Appendix B: Database Schema

B.1 Entity Relationship Diagram

User (1) $\longleftrightarrow$ (Many) UserProfile

User (1) $\longleftrightarrow$ (Many) Enrollment

User (1) $\longleftrightarrow$ (Many) Course (as teacher)

User (1) $\longleftrightarrow$ (Many) Evaluation (as student)

User (1) $\longleftrightarrow$ (Many) Evaluation (as evaluator)

Course (1) $\longleftrightarrow$ (Many) Video

Course (1) $\longleftrightarrow$ (Many) Note

Course (1) $\longleftrightarrow$ (Many) Quiz

Course (1) $\longleftrightarrow$ (Many) Evaluation

Quiz (1) $\longleftrightarrow$ (Many) Question

Quiz (1) $\longleftrightarrow$ (Many) QuizAttempt

Question (1) $\longleftrightarrow$ (Many) QuizAnswer